

Abstract--

The rapid evolution of big data analytics and machine learning (ML) has revolutionized traditional inventory management and demand forecasting in supply chain operations. This review examines the transformative impact of hybrid ML models, as exemplified by the LightGBM-PSO-LSTM framework proposed by Tang (2024), on optimizing inventory turnover, reducing forecasting errors, and enhancing operational efficiency. By integrating gradient boosting techniques with metaheuristic-optimized recurrent neural networks, such models address the limitations of conventional forecasting methods and enable data-driven, adaptive inventory control. This paper synthesizes key findings from Tang's study, highlighting the role of big data in facilitating real-time decision-making, minimizing costs, and improving service levels. It also discusses practical implications, challenges, and future directions for implementing intelligent inventory systems in modern supply chains.

Keywords: Big Data Analytics, Inventory Management, Demand Forecasting, LightGBM, PSO-LSTM, Supply Chain Optimization, Machine Learning.

1. Introduction

Supply chain management (SCM) has become increasingly complex due to globalization, fluctuating consumer demands, and the proliferation of multi-echelon distribution networks. Central to SCM is inventory management, which balances the dual objectives of minimizing holding costs and avoiding stockouts. Traditional inventory models, such as Economic Order Quantity (EOQ) and reorder point systems, often fail to account for nonlinear demand patterns and real-time variability. The advent of big data analytics and machine learning has introduced advanced predictive capabilities that enhance decision-making under uncertainty.

In their pioneering work, Tang (2024) demonstrated how big data analytics, combined with hybrid ML models, can significantly improve demand forecasting accuracy and inventory performance. This review paper critically examines Tang's methodology, results, and contributions to the field, contextualizing them within the broader landscape of intelligent supply chain systems.

2. Big Data Analytics in Supply Chain Inventory Management

Big data analytics enables organizations to process vast volumes of structured and unstructured data, uncovering insights that drive strategic inventory decisions. Tang (2024) emphasized the role of cloud-based platforms in facilitating high-performance data storage and computation, which are essential for real-time forecasting and inventory optimization.

Key benefits of big data in inventory management include:

- **Enhanced Forecasting Accuracy:** By analyzing historical sales, seasonality, market trends, and external factors, big data models generate more reliable demand forecasts.
- **Dynamic Inventory Control:** Real-time data integration allows for adaptive reorder points and safety stock levels.
- **Cost Reduction:** Improved forecasting reduces excess inventory and associated holding costs,

while also minimizing stockout-related losses.

Tang's study leverages these advantages to build a two-level multi-cycle inventory model that integrates forecasting outputs directly into inventory decision processes.

3. Hybrid Machine Learning Models for Demand Forecasting

3.1 LightGBM Model

LightGBM (Light Gradient Boosting Machine) is a high-efficiency gradient boosting framework that supports large-scale data processing with minimal computational overhead. Tang (2024) employed LightGBM for its ability to handle high-dimensional data and deliver fast, accurate predictions. The model's splitting mechanism, based on variance reduction, ensures optimal feature selection and robust performance.

3.2 PSO-LSTM Model

Long Short-Term Memory (LSTM) networks are well-suited for time-series forecasting due to their ability to capture long-term dependencies. However, LSTMs are sensitive to hyperparameter settings. Tang integrated Particle Swarm Optimization (PSO) to automate the tuning of LSTM parameters, enhancing model convergence and predictive accuracy. The PSO-LSTM hybrid effectively models nonlinear temporal patterns in demand data.

3.3 Combined LightGBM-PSO-LSTM Framework

Tang proposed a combined model that leverages the strengths of both LightGBM and PSO-LSTM. Experimental results showed that the hybrid model achieved the lowest root mean square error (RMSE = 0.7838) and mean absolute error (MAE = 0.2028) among all tested configurations. The model's superior fit and generalization capability make it suitable for real-world demand forecasting applications.

4. Two-Level Multi-Cycle Inventory Management Model

Tang's inventory model incorporates forecasted demand into a two-echelon system where each level—such as warehouse and retailer—operates with independent yet coordinated replenishment policies. The model minimizes total inventory cost, which includes:

- Ordering Costs: Fixed and variable costs associated with placing orders.
- Holding Costs: Costs related to storing inventory over time.

The model also incorporates constraints to ensure service levels, such as non-negativity of inventory and lead-time demand fulfillment. By aligning inventory decisions with ML-driven forecasts, the system achieves a balance between cost efficiency and service reliability.

5. Empirical Validation and Performance Metrics

Tang validated the proposed framework using real data from Company H. Key performance indicators included:

- Inventory Turnover Ratio: Increased from 8.2 (2022) to a peak of 9.2 (2023).
- On-Time In-Full (OTIF) Achievement Rate: Improved from 97.9% (2022) to 99.3% (2023).

These results confirm that the LightGBM-PSO-LSTM model not only enhances forecasting precision but also translates into tangible operational improvements.

6. Comparative Analysis with Existing Approaches

Compared to single models (e.g., LSTM, Random Forest, Linear Regression) and other hybrid combinations (e.g., RF-PSO-LSTM, LR-PSO-LSTM), the LightGBM-PSO-LSTM framework demonstrated superior performance. This aligns with broader trends in the literature, where hybrid ML models consistently outperform standalone methods in handling demand volatility and seasonality.

7. Challenges and Future Directions

Despite its success, the implementation of hybrid ML models in inventory management faces several challenges:

- Data Quality and Integration: Reliable forecasts depend on clean, consistent, and comprehensive data.
- Computational Complexity: Combining multiple algorithms requires significant processing power and expertise.
- Interpretability: Complex models like LSTM and LightGBM can act as "black boxes," limiting transparency.

Future research should focus on:

- Developing explainable AI (XAI) techniques for ML-based inventory systems.
- Integrating reinforcement learning for autonomous inventory control.
- Scaling hybrid models for multi-echelon, global supply chains.

8. Conclusion

Tang's (2024) study exemplifies the transformative potential of big data analytics and hybrid machine learning models in supply chain inventory management. The LightGBM-PSO-LSTM framework offers a scalable, accurate, and practical solution for demand forecasting and inventory optimization. As supply chains continue to evolve in complexity and dynamism, the

integration of intelligent analytics will be crucial for achieving resilience, efficiency, and competitive advantage.

References

- [1] Tang, W. (2024). Improvement of Inventory Management and Demand Forecasting by Big Data Analytics in Supply Chain. *Applied Mathematics and Nonlinear Sciences*, 9(1), 1–15.
- [2] Namira, K., Labriji, H., & Lahmar, E. B. (2022). Decision Support Tool for Dynamic Inventory Management Using Machine Learning, Time Series and Combinatorial Optimization. *Procedia Computer Science*, 198, 423–428.
- [3] Khedr, A. M., & Rani, S. (2024). Enhancing Supply Chain Management with Deep Learning and Machine Learning Techniques: A Review. University of Sharjah.
- [4] Park, M.-J., Turner, O., & Becker, T. (2025). Inventory Optimization in Retail Supply Chains Using Deep Reinforcement Learning. *Advances in Management and Intelligent Technologies*, 1(3).
- [5] Siddiqui, M. Z. (2024). Optimizing Supply Chain Dynamics Using Machine Learning. Rochester Institute of Technology.
- [6] Saggi, M. K., & Jain, S. (2018). A Survey Towards an Integration of Big Data Analytics to Big Insights for Value-Creation. *Information Processing & Management*, 54(5), 758–790.